

Multi-Row Subquery in MySQL

Definition: A multi-row subquery returns more than one value (multiple rows). It is commonly used with **IN**, **ANY**, and **ALL** operators.

Example Tables

id	name	salary	dept_id
1	Arun	50000	10
2	Ravi	60000	20
3	Kumar	45000	10
4	Priya	70000	30

1. Using IN

Find employees who belong to departments having more than one employee.

```
SELECT *
FROM employees
WHERE dept_id IN (
    SELECT dept_id
    FROM employees
    GROUP BY dept_id
    HAVING COUNT(*) > 1
);
```

Inner query returns: 10. The outer query finds employees whose dept_id is 10.

2. Using ANY

Find employees whose salary is greater than any salary in department 10.

```
SELECT *
FROM employees
WHERE salary > ANY (
    SELECT salary
    FROM employees
    WHERE dept_id = 10
);
```

The subquery returns 50000 and 45000. **> ANY** means the salary only needs to be greater than at least one returned value.

3. Using ALL

Find employees whose salary is greater than all salaries in department 10.

```
SELECT *
FROM employees
WHERE salary > ALL (
    SELECT salary
    FROM employees
    WHERE dept_id = 10
);
```

> ALL means the salary must be greater than every value returned by the subquery.

Quick Reference

Operator	Meaning
IN	Matches any returned value
ANY	Condition is true for at least one value
ALL	Condition is true for every value